B2.2 Organelles and compartmentalization (HL)

NAME	DATE	TOTAL
		_____ / 30

About 30 minutes, or one section at a time. The number of marks shows how many separate points to make.

Section A: Recall

7 MARKS

1 STATE the role of the cristae of a mitochondrion. [B2.2.4] [1]

2 STATE where the enzymes of the Calvin cycle are found in a chloroplast. [B2.2.5] [1]

3 STATE one molecule that leaves the nucleus through the pores, and one that enters. [B2.2.6] [2]

4 NAME the protein that forms a coat around vesicles in receptor-mediated endocytosis. [B2.2.9] [1]

5 STATE two ways in which the Golgi apparatus modifies proteins. [B2.2.8] [2]

Section B: Understanding

11 MARKS

6 EXPLAIN how the small volume inside thylakoids helps a chloroplast make ATP. [B2.2.5] [2]

7 EXPLAIN why the nuclear envelope breaks into vesicles at the start of mitosis. [B2.2.6] [2]

8 DISTINGUISH between the proteins made by free ribosomes and by ribosomes on the rough ER. [B2.2.7] [2]

9 **OUTLINE** how a protein is processed as it passes through the Golgi apparatus. [B2.2.8] [3]

10 **EXPLAIN** how clathrin helps a vesicle form. [B2.2.9] [2]

Section C: Working with evidence

8 MARKS

All data in this section are invented for practice. The patterns are realistic.

11 [B2.2.7, B2.2.8, B2.2.9]

Pancreas cells that secrete digestive enzymes were given radioactive amino acids for 3 minutes, then non-radioactive ones. Samples were taken at intervals, and the percentage of the radioactivity in each structure was measured.

TIME AFTER START / MIN	ROUGH ER / %	GOLGI APPARATUS / %	SECRETORY VESICLES / %
3	86	6	2
20	43	41	9
120	4	14	70

a **IDENTIFY** where most of the radioactivity was after 3 minutes. [1]

b **DESCRIBE** the changes in radioactivity in the Golgi apparatus. [2]

c **DEDUCE** the route taken by the newly made proteins, using the data. [2]

d **EXPLAIN** why little radioactivity was found in free ribosomes or the cytoplasm. [1]

e **SUGGEST** two reasons why these cells were chosen for this experiment. [2]

Section D: Longer answer

4 MARKS

12 **EXPLAIN** how the structure of a mitochondrion is adapted for producing ATP. [B2.2.4] [4]